DIVYANSHU GOYAL

Sunnyvale, CA ⋄ (+1) 404-414-3762 ⋄ divyanshu.g25@gmail.com

LinkedIn ⋄ GitHub ⋄ Google Scholar

SUMMARY

Applied Scientist and multimodal researcher with 10 years at the intersection of frontier ML research and production engineering. At Adobe, trains and fine-tunes **vision-language models** at scale, designs novel **evaluation and alignment** methods (**5 patents**, peer-reviewed and arXiv publications), and led the AI powering **Adobe GenStudio for Performance Marketing** — now serving Fortune 500 clients including Microsoft and AT&T. Deep expertise across **VLM training**, **multimodal reasoning**, and **LLM pretraining & scaling laws**, paired with the systems engineering to deploy them in production.

EXPERIENCE

Applied Scientist – Adobe Unified Platform, San Jose June 2021 – Present

⋄ ***Vision-Language Model Training & Evaluation***

- Fine-tuned a **Gemma-3-12B vision-language model** with **LoRA** across **8× H100 GPUs** (DDP, bfloat16, Flash Attention 2 + Liger kernels) for large-scale marketing-quality assessment.
- Pioneered an **observable injection** technique that grounds VLM quality reasoning in **14+ computed pixel-level metrics** (sharpness, exposure, color balance); built an interpretable scorer over these signals reaching **Pearson r ≈ 0.88** at **MAE 0.77**, outperforming GPT-5.4 (MAE 1.34) and Claude Opus 4.7 (MAE 1.95) baselines.
- Developed the **MQCore evaluation benchmark** (inspired by DCLM CORE) and a fault-tolerant async annotation pipeline processing **100+ images/min** over the **700K-image KADIS** corpus.

⋄ ***Multimodal Research & Benchmarking***

- Authored **DistortBench** (CVPR FGVC13 Workshop 2026) — a **13.5K-question** diagnostic benchmark spanning **27 distortion types**, 6 perceptual categories, and 5 severity levels.
- Evaluated **18+ frontier VLMs** (GPT-5.4, Claude Sonnet 4.6, Qwen, InternVL, Gemma) with bootstrap confidence intervals and balanced-accuracy metrics, exposing low-level perception as a major weakness — top model **61.9%** vs. **65.7% human baseline**.
- Created a curiosity-driven **LLM-as-a-judge** framework for personalized creative evaluation using Bayesian surprise as a quantifiable novelty signal (arXiv 2025), achieving **23% F1** and **38% correlation** improvements over baselines.

⋄ ***Production Multimodal Systems & Agents***

- Architected the **Unified Brand Service** (90K+ LOC across 3 global regions) powering **Adobe GenStudio** — multimodal brand-DNA extraction, RAG-based retrieval, and GPT-4-Vision content validation.
- Built a **multimodal agent** that drives Adobe Photoshop from natural language via a **49-tool** observe-think-act loop with vision grounding and an iterative quality-critic refinement layer.
- Led the foundational prototype of **Adobe GenStudio for Performance Marketing** (launched Oct 2024), now serving Fortune 500 clients including Microsoft and AT&T; invented patented VLM pipelines for **cross-cultural image adaptation** and **constrained machine translation** (US Patents 2026).

Machine Learning Engineer – Adobe Inc., Bangalore July 2016 – August 2019

- Co-engineered Adobe's **ML inference platform** serving **3,500 QPS/node** at **0.3–0.6ms latency** (p99), using CPU flame-graph profiling, JMH micro-benchmarking, and lock-free architectures.
- Designed a GC-free asynchronous event processing pipeline using the LMAX Disruptor ring buffer pattern, decoupling logging from prediction threads to eliminate back-pressure under peak load.
- Built a custom **model lifecycle management** platform — versioned deployment, real-time monitoring, and staged rollout across dev, staging, and production environments.
- Co-authored Adobe Tech Blog on engineering high-throughput, low-latency ML services.

PUBLICATIONS & PATENTS

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment arXiv 2025
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari [paper]
Novel curiosity-driven framework that personalizes creative writing evaluation to individual judgments using LLM-as-a-judge, addressing the gap in subjective assessment capabilities of LLMs.

Cultural Adaptation of Images Using Generative AI US Patent App. 2026
A. Eppa, M. Anand, **D. Goyal** [patent]
Multi-stage pipeline where a vision-language model grounds source images as text, conditioned on region-specific cultural guidelines, and a text-to-image model renders culturally adapted output.

Guiding Language Translation with Translation Documents Using ML US Patent App. 2026
D. Goyal, A. Eppa, M. Anand [patent]
Structured approach to constrained machine translation by extracting guidelines into language-agnostic facets and computing facet-specific adherence scores for linguistic and stylistic alignment.

Parameter-Efficient Model Serving Systems US Patent App. 2025
Y. Wang, N. Vangala, M. Anand, **D. Goyal**, et al. [patent]
System for efficiently serving fine-tuned models by dynamically loading parameter-efficient layers into pre-loaded base models, minimizing network transfer costs and container startup time.

Brand-Aligned Content Generation with Structured Knowledge US Patent App. 2025
M. Anand, J. Mathew, **D. Goyal** [patent]
Generative system that transforms unstructured brand information into structured brand DNA elements with confidence scoring, enabling brand-aligned content creation across channels.

SELECTED PROJECTS & TECHNICAL WRITING

VibeNanoChat — GPT-2 Scale LLM Pretraining Framework 2026
From-scratch decoder-only transformer (RoPE, RMSNorm, sliding-window attention) with one-knob depth-parameterized scaling, a custom distributed **Muon** optimizer with ZeRO-2 state sharding, and Flash-Attention-3 / bfloat16 training on 8× H100. Empirically verified the $\mathbf{C = 6ND}$ compute law across 40+ scaling-law runs. [code]

A Lazy Engineer's Guide to Scaling Transformers March 2026
Deriving all transformer architecture dimensions, training hyperparameters, and compute schedules from a single integer (depth), enabling trivial multi-scale training sweeps. [blog]

Going Fast: Every Optimization That Made LLM Training Fly March 2026
Six concrete optimizations — TF32, BF16 mixed precision, torch.compile, Flash Attention, parallel DataLoaders, and pinned memory — that collectively drove significant MFU improvements during GPT-2 scale training. [blog]

TECHNICAL SKILLS

Research:	LLM Pretraining & Alignment (SFT, RLHF, DPO), Vision-Language Models, Multimodal Agents, Evaluation & Benchmarking, Data Curation
Training & Serving:	PyTorch, Hugging Face Transformers/PEFT, DeepSpeed, FSDP, vLLM, Flash Attention, LoRA/QLoRA, CUDA, Weights & Biases
Infrastructure:	Docker, LangChain, RAG pipelines
Programming:	Python, C++, JavaScript

EDUCATION

Georgia Institute of Technology – M.S. Computer Science (Machine Learning) 2019 – 2021 | **GPA: 4.0**
BITS Pilani – B.E. Computer Science, M.Sc. Mathematics 2011 – 2016 | **GPA: 8.75/10**